

Poisson Distribution

- Type: Discrete.
- Notation: $\text{Poi} \sim (x; \lambda)$
- Parameter: only one that is λ
- $f(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad x = 0, 1, 2, \dots$
- $E(x) = \text{Mean} = \lambda$
- $V(x) = \sigma^2 = \lambda$
- Standard deviation = $\sigma = \sqrt{\lambda}$

Example

During a laboratory Experiment the Average number of radio active particles passing through a Counter in 1 millisecond is 4 what is the probability that 6 particles enter the Counter in a given millisecond.

Solution $x = 6, \lambda t = 4 \times 1 = 4$

$$\text{Poi} \sim (x; \lambda) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \Rightarrow (6; 4) = \frac{e^{-4} 4^6}{6!}$$

$$P(x=6) = \frac{e^{-4} \cdot 4^6}{6!} = 0.1041$$

$$\underline{P(x=6) = 0.1041}$$

Ex Ten is the average number of oil Tankers arriving each day at a certain port city. the facilities at the port can handle at most 15 tanker per day. what is the probability that on a given day tankers have to be turned away.

let X be the r.v for number of tankers.

$$K = 15, \quad \lambda = 10.$$

$$Poi \sim (K: \lambda) = \frac{e^{-\lambda} \cdot \lambda^K}{K!}$$

$$P(X > 15) = ?$$

$$P(X > 15) = 1 - P(X \leq 15)$$

$$= 1 - \sum_{K=0}^{15} P(K: 10)$$

$$\sum_{K=0}^{15} = \frac{e^{-\lambda} \cdot \lambda^K}{K!} \quad K = 0, 1, 2, 3, 4, 5, 6, 7, \dots, 15$$

$$P(X > 15) = 1 - 0.9513$$

$$\underline{P(X > 15) = 0.0487}$$

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Date _____

5.53 An Inventory Study determines that, on

Average, demands for a particular item
at a warehouse are made 5 times per day

What is the probability that on a given day

this item is requested

a) more than 5 times or $P(X > 5)$

$$\lambda = 5$$

$$P(X > 5) = 1 - P(X \leq 5)$$

$$= 1 - \sum_{x=0}^5 P(X=5)$$

$$= 1 - \sum_{x=0}^5 \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad x = 0, 1, 2, 3, 4, 5$$

$$P(X > 5) = 0.3840$$

b) not at all or $P(X=0)$

$$P(X=0) = P(0; 5)$$

$$P(X=0) = \frac{e^{-5} \cdot 5^0}{0!}$$

$$P(X=0) = e^{-5}$$

$$P(X=0) = 0.0067$$

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Date _____

5.89 A Secretary makes 2 errors per page on Average. What is the probability that on the next page he or she will make

a) 4 or more errors. $\lambda = 2$

$$P(X \geq 4) = 1 - P(X < 4)$$

$$P(X < 4) = \sum_{x=0}^3 \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$P(X \geq 4) = 1 - \sum_{x=0}^3 \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$P(X \geq 4) = 1 - \sum_{x=0}^3 \frac{e^{-2} \cdot 2^x}{x!} \quad x = 0, 1, 2$$

$$P(X \geq 4) = 0.1429$$

b) no errors. $\lambda = 2$

$$P(X=0) \sim P(0:2)$$

$$= \frac{e^{-2} \cdot 2^0}{0!}$$

$$P(X=3) = 0.1353$$

S-60. A Certain area of the USA is on Average hit by 6 hurricanes a year. find the probability that for a given year area will be hit by

a) fewer than 4 hurricanes or $P(X < 4)$

$$\begin{aligned} P(X < 4) &= \sum_{x=0}^3 P(X; 6) \quad \lambda = 6 \\ &= \sum_{x=0}^3 \frac{e^{-6} \cdot 6^x}{x!} \quad x = 0, 1, 2, 3 \\ &= 0.1512 \end{aligned}$$

b) any where from 6 to 8 hurricanes.

$$\begin{aligned} P(6 \leq X \leq 8) &= \sum_{x=6}^8 P(X; 6) \\ &= \sum_{x=6}^8 \frac{e^{-6} \cdot 6^x}{x!} \quad x = 6, 7, 8 \\ &= 0.4015 \end{aligned}$$

5.63 A restaurant chef prepares a tossed Salad Containing on Average 5 Vegetables. Find the probability that the Salad Contains more than 5 vegetables.

Solution $\lambda = 5$ By using Poisson

a) on a given day or $P(X > 5)$

$$P(X > 5) = 1 - P(X \leq 5)$$

$$= 1 - \sum_{x=0}^5 P(X=x)$$

$$= 1 - \sum_{x=0}^5 \frac{e^{-5} \cdot 5^x}{x!} \quad x=0,1,2,3,4,5$$

$$P(X > 5) = 0.3840$$

b) on 3 of the next 4 days

b) By using binomial $p = 0.3840$

$$b(3:4, 0.384) = \binom{4}{3} (0.384)^3 (0.616)$$

$$b(3:4, 0.384) = 0.1395$$

c) By using geometric (for the 1st time in April on 5 April)
 $p = 0.3840$

$$g(5:0.384) = (0.384)(0.616)^{5-1}$$

$$= (0.384)(0.616)^4$$

$$g(5:0.384) = 0.0553$$

5.26 the probability that a student fail the screening test is known to be 0.004 of the next 1875 students

find the probability that

a) fewer than 5 fail in Test or $P(X < 5)$

$$P(X < 5) = \sum_{x=0}^4 P(X=x)$$

$$n = 1875, p = 0.004$$

$$b \sim p \Rightarrow np = \lambda$$

$$(1875)(0.004) = 7.5$$

$$\lambda = 7.5$$

$$P(X < 5) = \sum_{x=0}^4 P(X=7.5)$$

$$= \sum_{x=0}^4 \frac{e^{-7.5} \cdot 7.5^x}{x!}, x=0,1,2,3,4$$

$$P(X < 5) = 0.1321$$

b) 8, 9, or 10 fail in the Test.

$$P(8 \leq X \leq 10) = \sum_{x=8}^{10} P(X=7.5)$$

$$= \sum_{x=8}^{10} \frac{e^{-7.5} \cdot 7.5^x}{x!}, x=8,9,10$$

$$P(8 \leq X \leq 10) = 0.3376$$

c) Mean = $E(X) = np = \lambda = 7.5$

5.64 The probability that a person will die

from a certain RT infection is 0.002.

Find the probability that fewer than 5
of the next 2000 so infected will die.

Sol $p = 0.002$, $n = 2000$

$$b \sim p = np = \lambda \Rightarrow 2000 \times 0.002$$

$$\lambda = 4$$

$$P(X < 5) = \sum_{x=0}^4 P(X: \lambda)$$

$$= \sum_{x=0}^4 \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad x = 0, 1, 2, 3$$

$$P(X < 5) = 0.6288$$

5.65 Suppose that on Average 1 person in 1000
makes a numerical error in
preparing his or her income tax return.
If 10,000 forms are selected at random
find the probability that 6, 7, or 8 of the
forms contain an error.

Sol. $n = 10,000$, $p = 1/1000 = 0.001$

$$b \sim p$$

$$np = \lambda \Rightarrow (10,000)(0.001) = 10$$

$$P(6 \leq X \leq 8) = ? \quad \sum_{x=6}^8 (X: 10)$$

$$= \sum_{x=6}^8 \frac{e^{-10} \cdot 10^x}{x!} \quad x = 6, 7, 8$$

$$= 0.2657$$



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Date _____

Ex-73 the probability that a person die
when he contracts a virus infection
is 0.001 of the next 4000
What is the mean number that
will die.

Sol $p = 0.001, n = 4000$

$$P \sim b \Rightarrow \lambda = np$$

$$\lambda = 0.001 \times 4000$$

$$\underline{\lambda = 4}$$

Ex. In a Certain Industrial facility accident occur infrequently. It is known that the probability of an accident on any given day is 0.005 and accidents are independent.

- a) What is the probability that in any give day period of 400 days there will be an accident on one day.

Here $n \rightarrow$ large, $p =$ small, So

$$b \sim p \Rightarrow np = p \Rightarrow np = \lambda$$

$$\lambda = 0.005 \times 400 \Rightarrow \lambda = 2$$

So using Poisson approximation.

$$\begin{aligned} P(X=1) &= \frac{e^{-\lambda} \cdot \lambda^x}{x!} \\ &= \frac{e^{-2} \cdot 2^1}{1!} \\ &= e^{-2} \cdot 2 \end{aligned}$$

$$P(X=1) = 0.271$$

Ex. in a manufacturing process where glass products are produced, defective or bubbles occur. It is known that, on average 1 in every 1000 of these items produced has one or more bubbles.

What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles.

Solution Here $p = \text{small}$ $n = \text{large}$.

$$p = 1/1000 = 0.001$$

$$n = 8000$$

$$b \sim p \Rightarrow np = \lambda \Rightarrow 8000 \times 0.001 = \lambda$$

$$\lambda = 8$$

X is random variable for number of bubbles. $P(X \sim \lambda) \sim P(X: \lambda)$.

$$P(X \sim \lambda) = P(X < 7) \sim P(X: \lambda)$$

$$P(X < 7) = \sum_{x=0}^6 \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$P(X < 7) = 0.3134$$